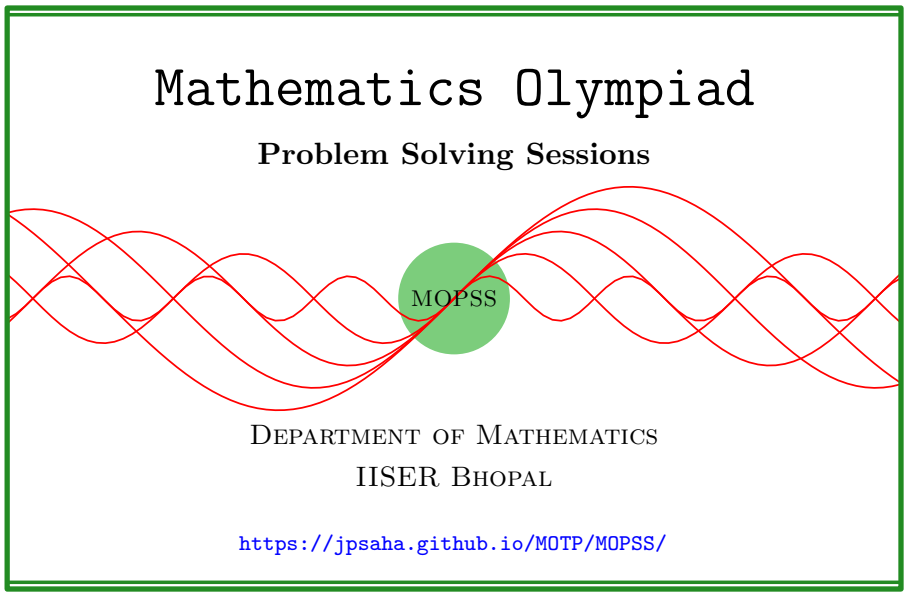
Telescoping

MOPSS



Suggested readings

- Evan Chen's advice On reading solutions, available at https://blog.evanchen.cc/2017/03/06/on-reading-solutions/.
- Evan Chen's Advice for writing proofs/Remarks on English, available at https://web.evanchen.cc/handouts/english/english.pdf.
- Notes on proofs by Evan Chen from OTIS Excerpts [**Che25**, Chapter 1].
- Tips for writing up solutions by Edward Barbeau, available at https://www.math.utoronto.ca/barbeau/writingup.pdf.
- Evan Chen discusses why math olympiads are a valuable experience for high schoolers in the post on Lessons from math olympiads, available at https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/.

List of problems and examples

2 Exercise (RMO 2018b P6,) 2

§1 Telescoping

Example 1 (IMOSL 1996 A7, Armenia)**.** Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that for all $x \in \mathbb{R}$, we have $|f(x)| \leq 1$ and

$$f\left(x + \frac{1}{6}\right) + f\left(x + \frac{1}{7}\right) = f(x) + f\left(x + \frac{13}{42}\right).$$

Show that f is periodic.

Solution 1. Consider the function $g : \mathbb{R} \to \mathbb{R}$, defined by $g(x) = f(x + 1/7) - f(x)$. Note that $g(x + 1/6) = g(x)$ holds for all $x \in \mathbb{R}$. Let $h : \mathbb{R} \to \mathbb{R}$ denote the function, defined by

$$h(x) = g(x) + g\left(x + \frac{1}{7}\right) + g\left(x + \frac{2}{7}\right) + \cdots + g\left(x + \frac{6}{7}\right).$$

Note that $h(x) = f(x + 1) - f(x)$ and $h(x + 1/6) = h(x)$ holds for all $x \in \mathbb{R}$. For any $x \in \mathbb{R}$ and any integer $r \geq 1$, we have

$$f(x + r) - f(x) = h(x) + h(x + 1) + \cdots + h(x + r - 1) = rh(x),$$

and this shows that $|rh(x)| \leq 2$. This implies that h is the zero function, and hence, f is periodic. ■

Exercise 2 (RMO 2018b P6,)**.** Define a sequence $\{a_n\}_{n\geq 1}$ of real numbers by

$$a_1 = 2, \qquad a_{n+1} = \frac{a_n^2 + 1}{2}, \text{ for } n \geq 1.$$

Prove that

$$\sum_{j=1}^{N} \frac{1}{a_j + 1} < 1$$

for every natural number N.

Solution 2. Note that

$$2(a_{n+1} - 1) = a_n^2 - 1$$

holds for any $n \geq 1$. Also note that $a_1 > 1$, and by induction, it follows that $a_n > 1$ for any $n \geq 2$. This shows that

$$\frac{1}{a_n + 1} = \frac{a_n + 1 - 2}{a_n^2 - 1}$$

$$= \frac{1}{a_n - 1} - \frac{2}{a_n^2 - 1}$$
$$= \frac{1}{a_n - 1} - \frac{1}{a_{n+1} - 1}$$

holds for any $n \geq 1$. Consequently, for any natural number N, we obtain

$$\sum_{j=1}^{N} \frac{1}{a_j + 1} = \sum_{j=1}^{N} \left(\frac{1}{a_n - 1} - \frac{1}{a_{n+1} - 1} \right) = \frac{1}{a_1 - 1} - \frac{1}{a_{N+1} - 1} < \frac{1}{a_1 - 1} = 1.$$

■

References

[**Che25**] Evan Chen. *The OTIS Excerpts*. Available at https://web.evanchen.cc/excerpts.html. 2025, pp. vi+289 (cited p. 1)